

Transformative Therapies for Depression: Postpartum Depression, Major Depressive Disorder, and Treatment-Resistant Depression

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Keywords

postpartum depression, major depression, treatment-resistant depression, neurosteroids, brexanolone, zuranolone, psilocybin

Abstract

Depressive disorders present an enormous global public health burden. A notable treatment gap exists between the prevalence of depression and our ability to provide rapid-acting, effective treatment that achieves remission. Brexanolone and zuranolone, the first US Food and Drug Administration–approved drugs for postpartum depression, signify a critical advancement in addressing the unmet needs of a vulnerable patient population. Psilocybin shows promise for treatment-resistant depression and for those who have struggled to find relief with existing treatments. This review discusses transformative therapies that represent significant advancements in postpartum depression, major depressive disorder, and treatment-resistant depression.

INTRODUCTION

Depressive disorders present an enormous global public health burden, are a major cause of morbidity and mortality, and by 2030 are expected to rank first in disease burden by the World Health Organization (1). The prevalence of depression has been growing over time and has markedly increased after the global COVID-19 pandemic (2). In 2019, prevalence data from the Centers for Disease Control estimated that ~18% of the population reported currently having depression, and 29% reported having a lifetime history of depression (3). Moreover, the lifetime prevalence of depression in women is twice that of men (4). A major reason for the increased prevalence in women is due to the high prevalence of reproductive mood disorders (5, 6).

Evidence-based recommendations for depressive disorders most commonly include psychotherapy, pharmacologic treatment, and neuromodulation therapies [electroconvulsive therapy (ECT), repetitive transcranial magnetic stimulation (rTMS), esketamine] (7). Lifestyle interventions to address sleep, physical activity, nutrition, and social support are also important aspects of a comprehensive treatment plan (7). A significant gap exists between the number of people who suffer from depression and our ability to provide effective treatment that achieves remission of illness (8). Therefore, there is a great need for transformation in the treatment of depressive disorders.

In this review, we discuss recent innovative work in the treatment of depression, with a specific focus on rapidly acting antidepressant pharmacologic agents with novel mechanisms of action. We begin by discussing the first US Food and Drug Administration (FDA)-approved medications for postpartum depression (PPD), the neurosteroid therapeutic agents brexanolone and zuranolone (9, 10). We subsequently discuss the treatment of major depressive disorder (MDD) and treatment-resistant depression (TRD) using psilocybin with assisted therapy, including efficacy, duration of effect, predictors of response, and study limitations.

BACKGROUND ON POSTPARTUM DEPRESSION

PPD is one of the most common complications of childbirth, impacting one out of seven women (11, 12). PPD is conceptualized as a major depressive episode in the perinatal period (onset during pregnancy or within the first 4 weeks postpartum) with core features of depression and themes related to parturition (13). Depressive episodes require at least five symptoms that have been present for 2 or more weeks and interfere with functioning (13). These symptoms may include depressed mood; anhedonia; feelings of guilt; changes in energy, concentration, appetite, and sleep; and suicidal ideation (13). Common themes may include difficulty with role transitions, poor attachment or bonding, regret, isolation, anxiety, and fears around infant and maternal health (9, 14–16).

PPD has historically been treated with both psychotherapy and pharmacologic therapies despite these pharmacological treatments not having indications specific for PPD (12). While treatable, there is strong evidence that PPD is underdiagnosed and undertreated (17). Cox et al. (17) did a systematic search to identify articles focused on diagnostic, treatment, and remission rates for perinatal depression. They found that only 30.8% of women were recognized as having PPD, only 15.8% received treatment, only 6.3% received adequate treatment, and only 3.2% achieved remission (17). These are tragic statistics for such a common experience.

There is clearly more work that needs to be done in reproductive psychiatry to improve outcomes for patients who suffer from PPD. The impact of untreated PPD is huge and multi-generational (15, 16, 18–21). Untreated perinatal mood and anxiety disorders can have maternal effects such as increased risk for preeclampsia, diabetes, and high blood pressure (22). The effects on the infant can include preterm delivery and low birth weight, not to mention impaired bonding

and attachment and breastfeeding difficulties (21, 23, 24). There is a high mortality rate in this population, with suicide being the highest cause of death for women up to 1 year after delivery (25). Thus, there is a need for rapidly acting antidepressants during this critical period.

In this section, we review allopregnanolone, an intrinsic hormone, and its relevance to PPD. This is followed by a discussion of the development of brexanolone and zuranolone, the first two FDA-approved medications for PPD (9, 10). Lastly, we review the current limitations of these neurosteroid therapeutics and provide a comparison table for clinicians.

Brief History of Allopregnanolone

An exemplary case of translational research is found in the development of allopregnanolone-based treatments for postpartum depression (26). In 1930, researchers first noticed the progestational qualities of a molecule produced in the adrenal gland (27). Building upon this discovery, in 1938, Beall and Reichstein (27) successfully isolated allopregnanolone. In 1981, Baulieu observed allopregnanolone production in the brain, expanding our understanding beyond its adrenal origin (26). By 1986, allopregnanolone was identified as a positive allosteric modulator of synaptic and extrasynaptic GABA_A receptors (26).

Further insights emerged in 1996 when research showed that rodent fluoxetine administration increased allopregnanolone levels in the brain, suggesting its potential anxiolytic or antidepressant effects (28). Building upon this foundation, in 1998, Concas's laboratory identified a correlation between changes in GABA_A receptor subunits and allopregnanolone fluctuations in the brain during pregnancy and parturition (29). Notably, a sharp decline in allopregnanolone levels following parturition was observed. Preclinical data highlighted the rapid fluctuations in allopregnanolone levels during the transition from pregnancy to the postpartum period, suggesting a potential trigger for postpartum depression in susceptible individuals (30, 31).

These discoveries paved the way for preclinical animal studies (32), which provided the rationale for the initial open-label trial of brexanolone, a proprietary formulation of allopregnanolone, for women suffering from postpartum depression.

Clinical Trials of Brexanolone

In 2015, brexanolone (United States Adopted Name; formerly SAGE-547 injection), a proprietary, intravenous form of allopregnanolone, was used in an open-label study to treat severe PPD (32). All four of the women enrolled had PPD [17-item Hamilton Depression Rating Scale (HAM-D) score ≥ 20] with the onset of symptoms in the third trimester or within 12 weeks of delivery. Brexanolone was titrated over 12 h to reach third-trimester allopregnanolone levels, held at a 36-h maintenance dose, and then tapered over the last 12 h for a total 60-h infusion (32). The University of North Carolina's Perinatal Psychiatry Inpatient Unit served as the site for this open-label trial, and all four women tolerated the infusion without serious adverse events. All four women achieved remission (HAM-D score ≤ 7) by hour 24.

The phase II double-blind randomized placebo-controlled trial (NCT02614547) maintained the same protocol (32). Twenty-one women with severe PPD (HAM-D score ≥ 26) were recruited, 10 received brexanolone, and 11 received a placebo. At the 60-h primary endpoint, the HAM-D mean reduction from baseline was 21.0 points [standard error (SE), 2.9] in the brexanolone group and 8.8 (SE, 2.8) in the placebo group [difference, -12.2 ; 95% confidence interval (CI), -20.77 to -3.67 ; $p = 0.0075$; effect size = 1.2].

The phase III trials consisted of two double-blind randomized placebo-controlled trials conducted at more than 30 clinical sites in the United States [NCT02942004 (study 1) and NCT02942017 (study 2)] (33). Those enrolled had severe PPD (HAM-D score ≥ 26) for one

of the phase III trials and a HAM-D score of 20–25 for the second study. The study enrolled 138 women who were randomly assigned (1:1:1) to receive brexanolone 90 µg/kg/h (BRX90), brexanolone 60 µg/kg/h (BRX60), or a placebo (33). In the second study, 108 women were randomly assigned to receive BRX90 ($n = 54$) or a placebo ($n = 54$). In the first study, the least-squares mean reduction in HAM-D score was 17.7 points (SE, 1.2) in the BRX90 group, 19.5 (SE, 1.2) in the BRX60 group, and 14.0 (SE, 1.1) in the placebo group [-5.5 (95% CI, -8.8 to -2.2); $p = 0.0013$ in the BRX60 group and 3.7 (95% CI, -6.9 to -0.5); $p = 0.0252$ in the BRX90 group]. At the 60-h primary endpoint in the second study, the least-squares mean reduction in HAM-D score was 14.6 points (SE, 0.8) in the BRX90 group and 12.1 (SE, 0.8) in the placebo group [-2.5 (95% CI, -4.5 to -0.5); $p = 0.0160$]. The most common adverse events in the brexanolone groups were headache, dizziness, and sleepiness (33).

In sum, the positive findings in the clinical trials supported the FDA approval of brexanolone in 2019 (9). The accelerated development timeline of the first approved drug for PPD and the first in its class was an important step forward for the field of reproductive psychiatry.

Clinical Trials of Zuranolone

Zuranolone, initially identified as SAGE-217, is an orally bioavailable positive allosteric modulator targeting both the extrasynaptic and intrasynaptic GABA_A receptors (34). Zuranolone underwent two pivotal clinical development programs: NEST for postpartum depression and LANDSCAPE for MDD (35). The NEST program comprised two significant clinical trials, ROBIN and SKYLARK (35).

ROBIN, a phase III double-blind randomized outpatient placebo-controlled clinical trial (NCT02978326), was conducted in 27 states to assess the efficacy and safety of zuranolone 30 mg. Severe PPD was defined as a HAM-D score of 26 or higher. All the women were less than 6 months postpartum. Patients in the trial were randomly assigned (1:1) to receive zuranolone or a placebo. The drug (zuranolone or placebo) was self-administered orally each evening for 2 weeks. All patients were followed until day 45. At the day 15 primary endpoint, the least-squares mean reduction in HAM-D score was 17.8 for zuranolone and 13.6 for the placebo [-4.2 (95% CI, -6.9 to -1.5); $p = 0.003$; effect size = 0.53]. The most commonly reported side effects in the zuranolone group were somnolence (15%), headache (9%), dizziness (8%), upper respiratory tract infection (8%), diarrhea (6%), and sedation (5%) (36). The clinical trial findings demonstrated efficacy and safety and supported further development of zuranolone as a potential treatment for PPD.

SKYLARK, a phase III double-blind randomized placebo-controlled clinical trial (NCT04442503), was conducted to assess the efficacy and safety of zuranolone 50 mg. Severe PPD was defined as a HAM-D score of 26 or higher, with the onset of symptoms beginning in the third trimester or less than 4 weeks postpartum. Patients in the trial were randomly assigned (1:1) in a stratified manner based on antidepressant use to receive either zuranolone or a placebo. The drug (zuranolone or placebo) was self-administered orally each evening for 2 weeks. All patients were followed until day 45. At the day 15 primary endpoint assessment, the HAM-D mean reduction from baseline was 15.6 (SE, 0.82) for zuranolone and 11.6 (SE, 0.82) for the placebo [-4.0 (95% CI, -6.3 to -1.7); $p = 0.001$; Cohen's $d = 0.52$]. The most commonly reported side effects in the zuranolone group were somnolence (26.5%), headache (9.2%), dizziness (13.3%), COVID-19 (5.1%), diarrhea (6.1%), nausea (5.1%), urinary tract infection (5.1%), and sedation (11.2%) (37). In sum, the phase III clinical trials supported the FDA's approval of zuranolone, a neurosteroid therapeutic agent, in August 2023 as the first oral drug for postpartum depression (10).

Limitations of Brexanolone and Zuranolone

In general, brexanolone is an effective and fast-acting antidepressant for PPD (9), but it is still not widely available due to the complexity of administration. As an intravenous infusion, the drug requires a risk evaluation and mitigation strategy (REMS) (38) and administration in a medical facility with insurance approval. Follow-up studies have explored breastfeeding recommendations with brexanolone. In short, breastfeeding during infusion is not contraindicated but should involve an informed consent process with the patient. Very low amounts of brexanolone are excreted into breastmilk (1–2% relative infant dose) (38, 39), and there have been no reports of the effects of brexanolone on milk production. No data are available on brexanolone in breastfed infants. A study that looked at 12 healthy adult lactating mothers found that at one-and-a-half days after the infusion, more than 95% of women had levels at or lower than 10 ng/mL (39). In a recent paper published by Brown University (39), 12 out of 26 patients elected to breastfeed during the brexanolone infusion, and no adverse events were reported. Mothers should monitor their infants for sedation if they choose to breastfeed during the infusion (40).

Zuranolone became commercially available in 2023 and has limited breastfeeding data. A clinical lactation study from 14 women taking 30 mg of zuranolone indicated low levels in breast milk with a calculated maximum relative infant dose of 0.357% (41). There have been no reports of the effects of zuranolone on milk production. No data on zuranolone in breastfed infants are available. Mothers should monitor their infant for sedation if they choose to breastfeed, particularly for preterm or newborn infants (42). **Table 1** compares key characteristics of brexanolone and zuranolone.

PSILOCYBIN IN MAJOR DEPRESSIVE DISORDER AND TREATMENT-RESISTANT DEPRESSION

Background on Psilocybin in Major Depressive Disorder and Treatment-Resistant Depression

Novel treatments for depression with rapid onset of effects and a known duration of action are essential. A resurgence of interest in the partial 5-HT_{2A} agonist psilocybin with adjunctive psychotherapy followed two randomized controlled trials (RCTs) in cancer patients (44). In both studies, a single session of psilocybin with assisted therapy resulted in dramatic relief of anxiety and depression. The therapeutic effect appeared to last 5 to 6 weeks. The cancer patients ranked the single session of psilocybin with assisted therapy as one of the most meaningful experiences of their lives (44). This was the impetus for the FDA to subsequently grant “breakthrough status” to psilocybin (45).

Psilocybin with assisted therapy has been studied in a number of psychiatric disorders and could have a great impact on MDD and TRD. TRD occurs when patients with MDD do not respond to—or are intolerant of—two adequate oral antidepressant trials with medication from different pharmacologic classes. TRD is common: 60% of MDD patients do not respond sufficiently to their initial oral antidepressant treatment (46, 47). TRD response is short-lived. Only 67% of MDD patients undergoing four oral antidepressant trials in the Sequenced Treatment Alternatives to Relieve Depression study remained in remission after 14 months (47). Relapse in TRD is frequent: 10–20% of patients with MDD remained significantly symptomatic for 2 years or longer after sequential oral antidepressant treatment (48). Treatment options for TRD are limited: Intranasal esketamine or ketamine infusions, rTMS, and ECT are effective but expensive and time-consuming, have significant side effects, or carry significant risks. The remaining treatment options—vagal nerve and deep brain stimulation—require invasive neurosurgery and years of neurostimulator monitoring.

Table 1 Comparison of key characteristics of brexanolone versus zuranolone (10, 38, 41–43)

Brexanolone (Zulresso)	Zuranolone (Zurzuvae)
Rapid-acting, single-treatment course	Rapid-acting, single-treatment course
Positive allosteric modulator of the GABA _A receptor	Positive allosteric modulator of the GABA _A receptor
Synthetically identical form of allopregnanolone	Altered to be active and orally bioavailable
Manufactured by Sage Therapeutics	Manufactured by Sage Therapeutics and Biogen
FDA approved in 2019	FDA approved in 2023
15 years and older	18 years and older
Injection/intravenous	Capsule/oral (20 mg, 25 mg, and 30 mg)
60-h continuous infusion, half-life 9 h	Oral at-home once-daily dosing for 14 days, half-life ~25 h
Weight-based dosing, titrated up in 4-h increments	50-mg dosing for 14 days, reduced dosing available for central nervous system side effects; 30-mg dosing for severe hepatic or moderate-severe renal impairment
Monotherapy or as an adjunct to other antidepressant treatments	Monotherapy or as an adjunct to other antidepressant treatments
Requires hospitalization (prior authorization required) at an approved healthcare site	Self-administered (prior authorization often needed)
Enrollment in REMS due to boxed warning	Does not require REMS enrollment
Boxed warning includes excessive sedation and sudden loss of consciousness	Boxed warning includes impaired ability to drive or engage in other potentially hazardous activities
Avoid in end-stage renal disease (potential accumulation of the solubilizing agent)	Metabolized by the hepatic enzyme CYP3A4
No contraindications	No contraindications
Effective in 60 h	Effective at day 14
90-day follow-up data	45-day follow-up data
Lactation data: relative infant dose 1–2%	Lactation data: relative infant dose <1%

Abbreviations: FDA, US Food and Drug Administration; REMS, risk evaluation and mitigation strategies.

Clinical Trials of Psilocybin in Major Depressive Disorder

Davis et al. (49) studied 24 participants with MDD in a randomized, waitlist-controlled trial of two doses of psilocybin. Participants could not be on antidepressants or have a history of suicide attempts. Participants received psilocybin 20 mg/70 mg and then 30 mg/70 kg in two sessions (average about 1.8 weeks apart) immediately or were assigned to an 8-week waitlist. The primary outcome was a depression score on the GRID Hamilton Depression Rating Scale (GRID-HAMD). Participants had an average GRID-HAMD score of 22.8 (3.9) [mean (standard deviation)] at baseline. At 1 and 4 weeks (the equivalent of 5 and 8 weeks for the waitlist group), participants in the immediate treatment group had average GRID-HAMD scores of 8.0 (7.1) and 8.5 (5.7) compared with participants in the waitlist group who had average GRID-HAMD scores of 23.8 (5.4) and 23.5 (6.0). Seventy-one percent of participants had a clinically significant response (decrease in GRID-HAMD of ≥50%) at weeks 1 and 4. At weeks 1 and 4, 58% and 54% of participants, respectively, had remission (GRID-HAMD <7). Following these 24 participants longitudinally, in a study by Gukasyan et al. (50), 75% of participants showed a response (decrease in GRID-HAMD of ≥50%), and 58% of participants experienced remission (GRID-HAMD <7) at 12 months after treatment.

Raison et al. (51) studied one dose of psilocybin 25 mg compared with niacin 100 mg in 104 participants with MDD. The primary outcome was a change in depression on the

Montgomery-Åsberg Depression Rating Scale (MADRS) from baseline to day 43 after treatment. Participants were not on psychotropic medications or receiving outside therapy during the study. On day 43, participants in the psilocybin group had a significantly greater reduction in MADRS score [-12.3 (95% CI, -17.5 to -7.2)]. Forty-one percent of participants receiving psilocybin had a sustained response compared with 11.4% in the niacin group.

Carhart-Harris et al. (52) studied 59 participants with MDD in a double-blind RCT comparing psilocybin and escitalopram. Participants stopped psychiatric medications at least 2 weeks before starting the study. Participants received psilocybin 25 mg 3 weeks apart with daily placebo or psilocybin 1 mg 3 weeks apart with daily escitalopram. The primary outcome was a change in depression score on the 16-item Quick Inventory of Depression Symptomatology Self-Report (QIDS-SR16). Change in depression scores did not differ significantly between groups for the primary outcome, with both groups showing a reduction in QIDS-SR16 from baseline to week 6 follow-up. Seventy percent of participants in the psilocybin group, compared with 45% in the escitalopram group, showed a response at 6 weeks (decrease in QIDS-SR16 of $\geq 50\%$). Fifty-seven percent in the psilocybin group, compared with 28% in the escitalopram group, experienced remission at 6 weeks (QIDS-SR16 ≤ 5).

Clinical Trials of Psilocybin in Treatment-Resistant Depression

Carhart-Harris et al. (53) studied participants with TRD, initially enrolling 12 participants in an open-label trial of two doses of psilocybin, 10 mg and then 25 mg, about 7 days apart. Participants in the study had moderate-to-severe depression (HAM-D ≥ 17) and failed to respond to adequate trials (6 or more weeks) of two antidepressants from different classes. The primary outcome was a change in QIDS-SR16 from baseline to 1 week after psilocybin 25 mg. One week after the psilocybin 25 mg treatment, the average decrease in QIDS-SR16 was 11.8 (95% CI, -9.15 to -14.35). At 3 months after treatment, the QIDS score remained significantly reduced [mean, -9.2 (95% CI, -5.69 to -12.71)].

An additional eight participants were subsequently included in the study by Carhart-Harris et al. (54) for a total of 20 participants with TRD who were followed for 6 months after treatment with psilocybin 10 mg and 25 mg. The primary outcome was depression on the QIDS-SR16 at 5 weeks after treatment, with depression also measured at 1, 2, and 3 weeks and 3 and 6 months after treatment. Depression scores decreased at all follow-up points, with maximum reduction at week 5. All participants had a reduction in depression score at week 1, with a maximum reduction at week 5.

In a phase II RCT sponsored by Compass Pathways (55), Goodwin et al. (56) studied 233 participants with TRD, randomizing them to receive one dose of psilocybin at either 25 mg, 10 mg, or 1 mg (control). The primary outcome was a change in depression on the MADRS between baseline and week 3 after treatment. About two-thirds of participants were on antidepressants at screening; antidepressants and other prohibited medications were tapered off at least 2 weeks before dosing. There was a significant difference between the change in MADRS for the 25 mg group compared with the 1 mg group at the 3-week follow-up [-6.6 (95% CI, -10.2 to -2.9 ; $p < 0.001$)]. There was not a statistically significant difference in change in depression score from baseline to 3 weeks for the 10 mg group compared with the 1 mg group. At the 3-week follow-up for the 25 mg group, 37% showed a response ($\geq 50\%$ decrease from baseline in MADRS total score) and 29% showed remission (MADRS total score ≤ 10), compared with 18% and 8%, respectively, in the 1 mg group. The difference in response and remission at 3 weeks for the 1 mg compared with the 25 mg group was statistically significant. However, sustained response at 12 weeks was not statistically different between the groups.

Psilocybin with Selective Serotonin Reuptake Inhibitors

RCTs demonstrated the acute antidepressant effect of psilocybin with assisted therapy in MDD, not only as monotherapy but also as an adjunctive treatment to selective serotonin reuptake inhibitors (SSRIs) (52). Goodwin et al. (57) examined psilocybin in 19 participants with TRD taking a concomitant SSRI medication. In this open-label study, the primary efficacy endpoint was a change in depression score (i.e., MADRS) from baseline to week 3. At 3 weeks, the mean change in MADRS was -14.9 (95% CI, -20.7 to -9.2), and 42% of participants ($n = 8$) had achieved both response and remission. There were no serious adverse events. Though further studies must be done, this suggests that psilocybin treatment while staying on antidepressant medications could be safe and effective.

One Versus Two Psilocybin Treatments

Investigating one versus two initial psilocybin-assisted therapy treatments for both short- and long-term outcomes is important in determining future standards of care. Prior studies of psilocybin-assisted therapy for the treatment of depression have used either one dose (55) or two doses (49, 52, 53). Future research should assess the acute effects of one versus two psilocybin-assisted therapy treatments, the response and remission rates, and the duration of response.

Duration of Psilocybin's Antidepressant Effect

The duration of psilocybin's antidepressant effect is not known. Despite the high relapse rate of depression in MDD and TRD, there are few published studies on the duration of the effect of psilocybin with assisted therapy in depression (50, 52, 54). As previously mentioned, Davis et al. (49) initiated a randomized trial of psilocybin (without other antidepressant medications) in MDD and reported on their follow-up of patients 12 months after treatment (50). Treatment response ($\geq 50\%$ decrease in HAM-D score from baseline) and remission (HAM-D ≤ 7) at 1 year were 75% and 58%, respectively. Before entry into this study, all participants stopped their antidepressants. At the 12-month follow-up, 33% of participants resumed antidepressant medications. So, up to one-third of the subjects had worsening depressive symptoms severe enough to require antidepressant treatment.

As previously mentioned in the section titled Clinical Trials of Psilocybin in Treatment-Resistant Depression, a phase II RCT sponsored by Compass Pathways (55), Goodwin et al. (56) is the largest randomized controlled double-blind psilocybin-assisted therapy study for the treatment of TRD. The results showed rapid and sustained response for patients receiving a single dose of psilocybin with psychological support. Sustained response up to week 12 was observed in 20.3% of patients who received 25 mg in the Compass Pathways clinical trial. The acute antidepressant effect of psilocybin with assisted therapy in TRD was of 12 weeks duration in one-fifth of patients (55).

There is evidence in the existing literature that psilocybin's antidepressant effect wanes after 12 weeks or 3 months. For example, in the open-label study of 12 TRD patients conducted by Carhart-Harris et al. (53), psilocybin's rapid antidepressant effect was demonstrated 1 week after treatment by a mean 11.8-point decrease in QIDS scores and a mean 25-point decrease in Beck's Depression Inventory (BDI) scores (53). However, signs of potential relapse emerged 3 months later. At the 3-month follow-up, mean QIDS scores increased by 2.6 points (from 7.4 to 10.0, or 22% of the initial 11.8-point decrease), and mean BDI scores increased by 6.5 points (from 8.7 to 15.2, or 26% of the initial 25-point decrease). The investigators identified 7 additional MDD patients, added them to the original 12 patients, and followed the sample of 19 out to 3 and 6 months (54). Analysis of BDI scores in the 19 subjects showed a replication of psilocybin's acute

antidepressant effect. One week after acute treatment, the 19 showed a mean 22.7-point decrease in BDI scores. However, at the 3-month follow-up, the mean BDI of the 19 subjects increased by 7.4 points (from 11.8 to 19.6, or 32.6% of the 22.7-point decrease), which exceeded the 26% increase observed in the 12 subjects. Six months after treatment, there were signs of impending relapse. At the 6-month follow-up, the mean BDI scores of the 19 subjects increased by 7.8 points (from 11.8 to 19.5, or 34.3% of the initial 22.7-point decrease). Based on the 32.6% and 34.3% increase in mean BDI scores at 3 and 6 months, respectively, further investigation of relapse in TRD following psilocybin-assisted therapy treatment for depression is essential.

Predictors of a Positive Outcome

Several predictors of positive outcomes following psilocybin-assisted therapy have emerged. Griffiths et al. (44) found that mystical-type experiences on the psilocybin session day mediated the effect of psilocybin dose on therapeutic outcomes on depression and anxiety in cancer patients. Ross et al. (58) also found that psilocybin-induced mystical experience mediated the therapeutic effect of psilocybin on anxiety and depression in cancer patients. A mystical experience during psilocybin-assisted therapy was also a predictor for a positive outcome in short-term (49) and long-term follow-up (50) in patients with MDD. The Mystical Experience Questionnaire (MEQ-30) and its subscales used in these studies are established (59) and validated (60) measures used to characterize mystical experience during treatment in order to determine their correlation with outcomes. The therapeutic alliance is also a predictor for a positive outcome during psilocybin-assisted therapy for depression (61). The association between therapeutic alliance, as measured with the scores of the Therapeutic Alliance Scale, and response can be measured. In addition, patient-related factors such as expectations and psychological flexibility can be correlated with outcomes.

Limitations of Psilocybin Research

The evidence supporting the efficacy of psilocybin with assisted therapy in MDD and TRD is promising. Limitations of studies of psilocybin include small sample sizes, difficulty with blinding, expectancy bias, and excluding patients on antidepressants, as well as the use of one versus two doses. The duration of the antidepressant is unclear. Predictors for positive outcomes remain limited. In general, studies of psilocybin for MDD and TRD have excluded participants with a personal or family history of bipolar disorder or psychotic disorders, active suicidal ideation, and major health issues, including hypertension or arrhythmia. Studies have also excluded participants taking psychotropic medications due to potential interactions with psilocybin or a theoretical reduction of response to psilocybin. The most common psychological and physiological side effects experienced during psilocybin-assisted therapy dosing sessions include anxiety, psychological discomfort, transient paranoia, visual distortion, negative emotional state, transient headache, mild nausea, elevated blood pressure, and elevated heart rate (62). Elevations in blood pressure that occurred during sessions resolved without intervention. Other adverse events reported during sessions included mild, controllable muscle motion, tenseness and soreness, chest tightness, vivid dreams, altered body sensation, diarrhea, and sleep disorder (62). In the Compass Pathways study, one participant received rescue medication (lorazepam) for anxiety. There is a theoretical risk of persisting hallucination (e.g., hallucinogen persisting perception disorder) or prolonged psychosis, but these have not been reported in studies to date.

OVERALL CONCLUSIONS

This review discusses transformative therapies that represent significant advancements in treatment for PPD, MDD, and TRD. Brexanolone and zuranolone have demonstrated substantial and

rapid antidepressant treatment for women suffering from PPD, a condition with profound ramifications for maternal and infant health. Additionally, psilocybin has shown a significant and rapid reduction in depression. The duration of sustained improvement in depressive symptoms in both MDD and TRD will be clarified with further research. Similarly, predictors for positive outcomes will be elucidated with additional studies. In sum, the transformative therapies of brexanolone, zuranolone, and psilocybin discussed in this article hold tremendous promise for revolutionizing the treatment landscape of depression and improving the lives of those affected by depression.

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